

Exoplanets – Hands-on MCMC

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Some terminology

- We have N observations of a variable y (e.g., the flux of the star) taken at time t and with associated measurement uncertainties σ
- We also have a model m that predicts the value of the variable at a function of time and for a set of M parameters $\boldsymbol{\theta} = \{\theta_j\}_{j=1,\dots,M}$ (e.g., planetary radius, orbital period..) so that $m_i = m(t_i, \boldsymbol{\theta})$
- Definition of chi square:

$$\chi^2 = \sum_{n=1}^N \frac{(y_i - m_i)^2}{\sigma_i^2}$$

- $\mathbf{y} = \{y_i\}_{i=1,\dots,N}$ is also denoted as $\mathbf{D}$

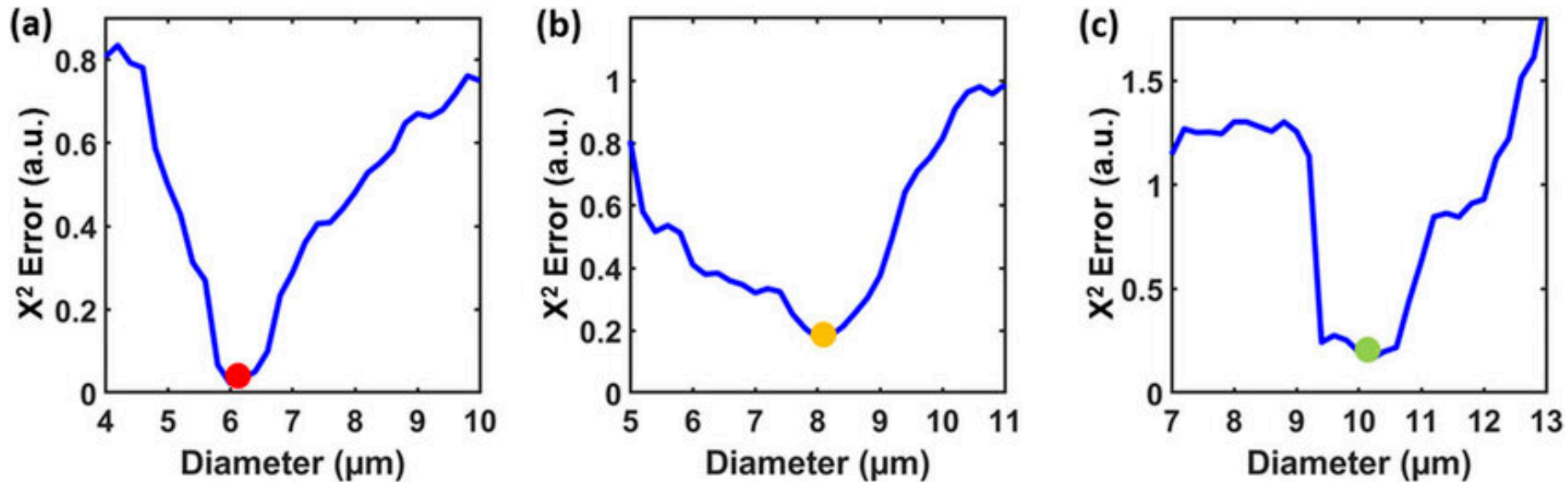
Frequentist vs Bayesian statistics

➤ Frequentist approach:

My variable x has *a true, exact* value

More measurements D give a higher probability of measuring that value

- Minimization of χ^2 to determine the variable value



- Standard tests or randomised sampling to determine the error
- Gaussian distributions everywhere $x = a \pm \sigma_a$

Frequentist vs Bayesian statistics

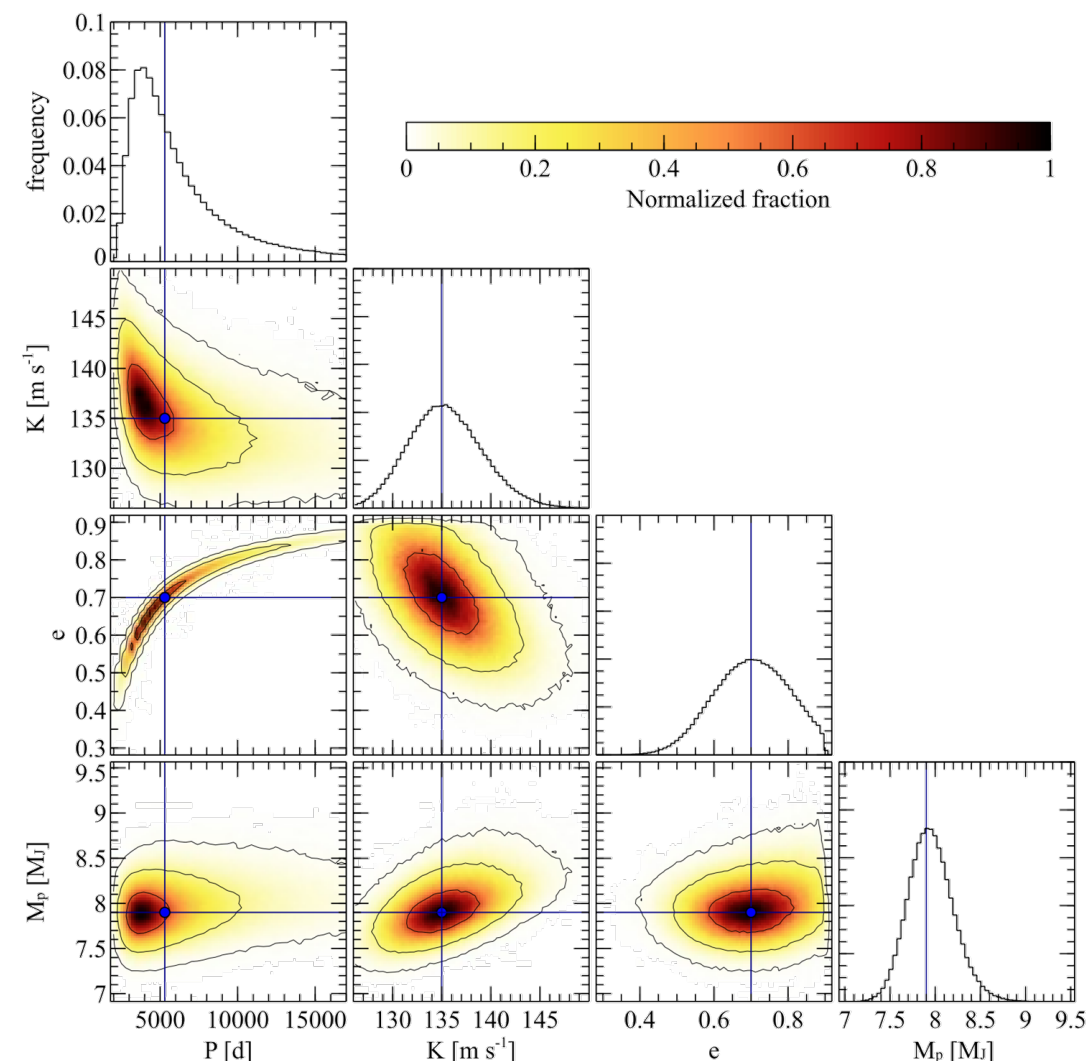
➤ Bayesian approach:

Probability of my *variable* of being within a given range of values

- *Prior*: distribution probability of my variable *before* performing my analysis (can be flat)
- *Posterior*: probability distribution of the variable *after* considering the data

The results of the analysis are the *posteriors*!
The values in the table are a simplification of these results

Parameter	Pr 0211c	Unit
P	5300^{+4450}_{-1800}	days
K	135 ± 4	m s^{-1}
e^a	0.70 ± 0.10	
$M_p \sin i^a$	7.95 ± 0.25	M_J



Bayesian formalism

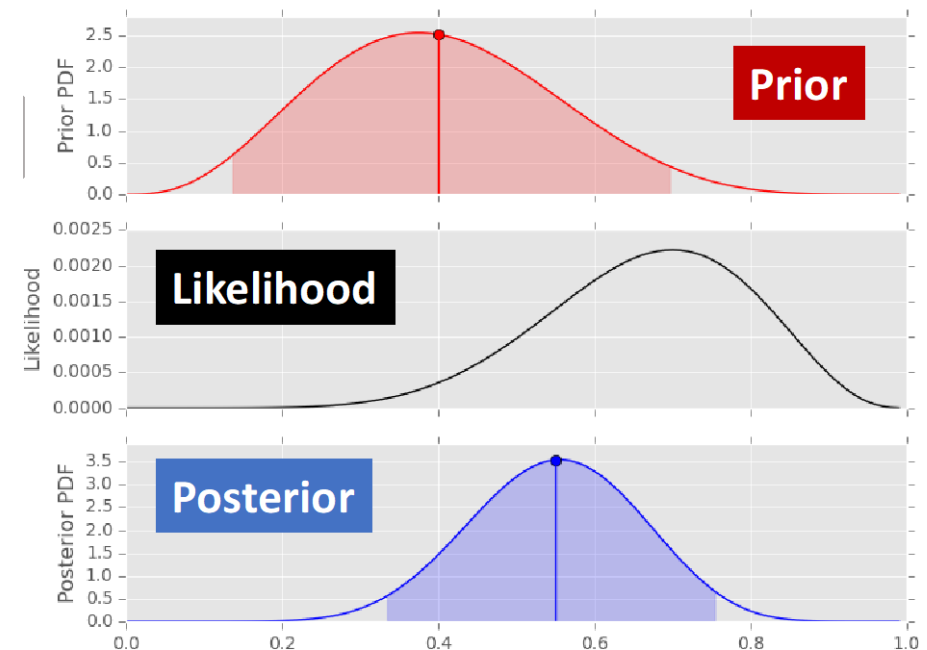
- **Prior** $P(\boldsymbol{\theta})$: previous knowledge on the model parameters (e.g., limb darkening coefficients from stellar atmosphere models)
- **Likelihood** $P(\mathbf{D}|\boldsymbol{\theta})$: probability of observing the data given a set of parameters
- **Evidence** $P(\mathbf{D}) = \int P(\mathbf{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})d\boldsymbol{\theta}$ (also called marginal likelihood): probability of observing the data under any possible value of the parameters
- **Posterior** $P(\boldsymbol{\theta}|\mathbf{D})$: probability of the model parameters given the data

Bayes' theorem:

$$P(\boldsymbol{\theta}|\mathbf{D}) = \frac{P(\mathbf{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\mathbf{D})}$$

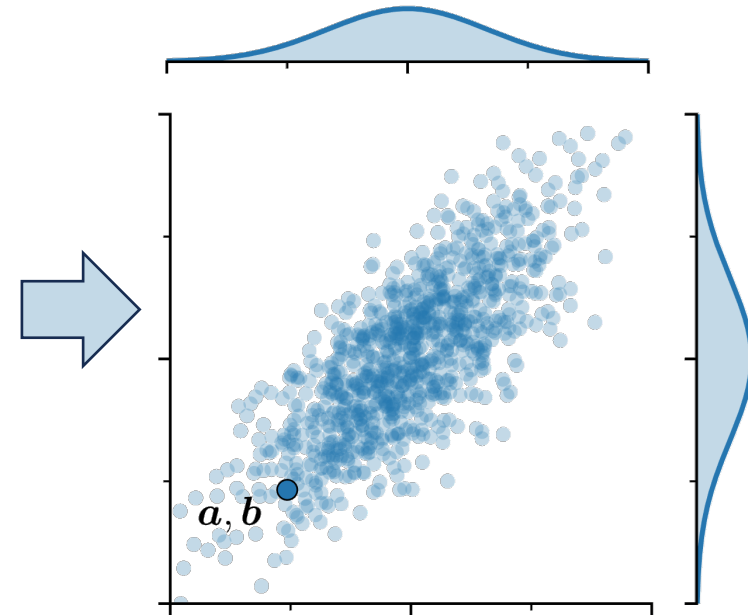
If $P(\mathbf{D})$ is constant (data is not changing through the analysis) then:

$$P(\boldsymbol{\theta}|\mathbf{D}) \propto P(\mathbf{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})$$



Approximating the posterior distribution

- **Evidence** $P(\mathbf{D}) = \int P(\mathbf{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})d\boldsymbol{\theta}$ nearly impossible to solve analytically
- The analytical solution of **Likelihood** $P(\mathbf{D}|\boldsymbol{\theta})$ * **Prior** $P(\boldsymbol{\theta})$ can be too complex, e.g, high-dimensional problems
- **Posterior** $P(\boldsymbol{\theta}|\mathbf{D})$ can be obtained by **sampling**:
 - A random set of $\boldsymbol{\theta}$ is generated
 - $P(\mathbf{D}|\boldsymbol{\theta})$ and $P(\boldsymbol{\theta})$ are evaluated to compute $P(\boldsymbol{\theta}|\mathbf{D})$
 - The process is repeated thousands/millions of time



Two preferred approaches (in astrophysics)

- Markov chain Monte Carlo (**MCMC**)
- Nested Sampling (**NS**)

But first: how can we compute Likelihood and Prior ?

Bayes' theorem

$$P(\boldsymbol{\theta}|\mathbf{D}) = \frac{P(\mathbf{D}|\boldsymbol{\theta})P(\boldsymbol{\theta})}{P(\mathbf{D})}$$

Computation of the likelihood (1)

- Using the terminology introduced in the first slide, each observation can be expressed as

$$y_i = m(t_i, \boldsymbol{\theta}) + \epsilon_i \quad \text{or simply } y_i = m_i + \epsilon_i$$

where ϵ_i is the (unknown) measurement error affecting our observation - do not confuse it with our *estimate* of the error σ_i)

- Let's assume that ϵ_i is drawn from a Normal distribution with mean 0 and variance σ_i^2 :

$$p(\epsilon_i) = \mathcal{N}(0, \sigma_i^2) \equiv \frac{1}{\sqrt{2\pi}\sigma_i} e^{\left(-\frac{\epsilon_i^2}{2\sigma_i^2}\right)}$$

- The value of y_i is equal to ϵ_i plus a constant equal to $m(t_i, \boldsymbol{\theta})$, so the probability distribution of y_i will be the same of ϵ_i , but with mean equal to m_i

$$p(y_i) = \mathcal{N}(m_i, \sigma_i^2) \equiv \frac{1}{\sqrt{2\pi}\sigma_i} e^{\left[-\frac{(y_i - m_i)^2}{2\sigma_i^2}\right]}$$

Computation of the likelihood (2)

- The probability of observing y_i assuming that the value is generated by the model m with parameters $\boldsymbol{\theta}$ for the epoch t_i by definition corresponds to the likelihood of the observation y_i :

$$\mathcal{L}_i(\boldsymbol{\theta}) \equiv p(y_i|\boldsymbol{\theta})$$

- If we assume that the noise is uncorrelated (*white* noise), then each ϵ_i is drawn from its corresponding distribution, independently from each other. The likelihood of the entire dataset is merely the products of the individual observations:

$$\mathcal{L}(\boldsymbol{\theta}) \equiv p(\mathbf{y}|\boldsymbol{\theta}) = \prod_{i=1}^N \mathcal{L}_i(\boldsymbol{\theta}) = \prod_{i=1}^N \left\{ \frac{1}{\sqrt{2\pi}\sigma_i} e^{\left[-\frac{(y_i-m_i)^2}{2\sigma_i^2}\right]} \right\}$$

- We can drop the $(\boldsymbol{\theta})$ dependency:

$$\mathcal{L}(\boldsymbol{\theta}) \equiv \mathcal{L}; \quad \mathcal{L}_i(\boldsymbol{\theta}) \equiv \mathcal{L}_i$$

Computation of the likelihood and posterior

➤ For numerical reasons, it is more convenient to compute the logarithm of $\mathcal{L}(\boldsymbol{\theta})$:

$$\ln \mathcal{L} = \sum_{i=1}^N \ln \mathcal{L}_i = \sum_{i=1}^N \ln \left(\frac{1}{\sqrt{2\pi}\sigma_i} e^{\left[-\frac{(y_i-m_i)^2}{2\sigma_i^2}\right]} \right)$$

➤ Keeping in mind the properties of the logarithm and the definition of χ^2 ;

$$\ln \mathcal{L} = \sum_{i=1}^N \ln \left(\frac{1}{\sqrt{2\pi}\sigma_i} \right) + \sum_{i=1}^N \left[-\frac{(y_i-m_i)^2}{2\sigma_i^2} \right]$$

$$\ln \mathcal{L} = \sum_{i=1}^N \ln \left(\frac{1}{\sqrt{2\pi}} \right) + \sum_{i=1}^N \ln \left(\frac{1}{\sigma_i} \right) - \frac{1}{2} \sum_{i=1}^N \left[\frac{(y_i-m_i)^2}{\sigma_i^2} \right]$$

$$\chi^2 = \sum_{n=1}^N \frac{(y_i - m_i)^2}{\sigma_i^2}$$

$$\ln \mathcal{L} = -0.5N \ln(2\pi) - \sum_{i=1}^N \ln(\sigma_i) - 0.5\chi^2$$

$$\Rightarrow \Rightarrow \quad \ln \mathcal{L} = -0.5 \left(N \ln 2\pi + \sum_{i=1}^N \ln \sigma_i^2 + \chi^2 \right)$$

➤ The logarithm of the prior can be added to the logarithm of the likelihood to obtain the posterior value for the given set of parameters $\boldsymbol{\theta}$:

$$\ln(P(\boldsymbol{\theta}|\mathbf{D})) \propto \ln(P(\mathbf{D}|\boldsymbol{\theta})) + \ln(P(\boldsymbol{\theta}))$$

More general form

Uncorrelated (white) noise:

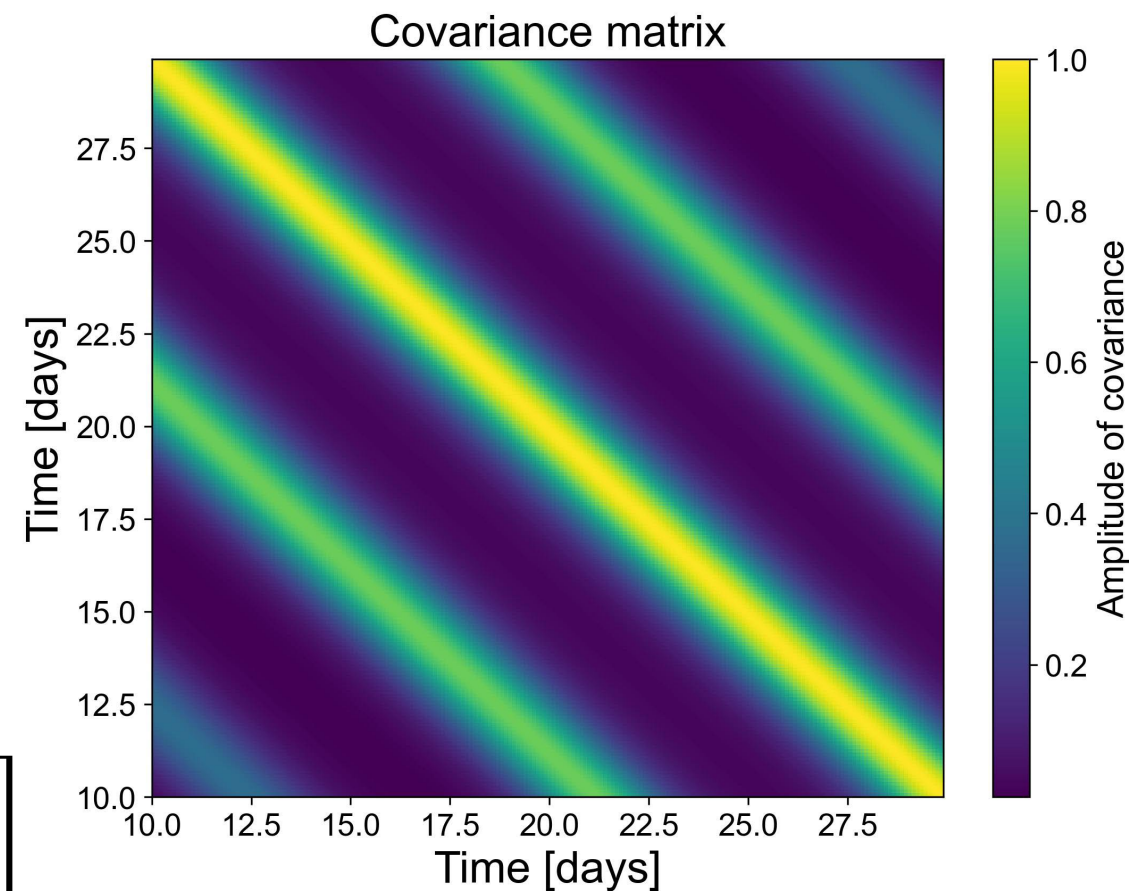
$$\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^N \left\{ \frac{1}{\sqrt{2\pi}\sigma_i} \exp \left[-\frac{(y_i - m_i)^2}{2\sigma_i^2} \right] \right\}$$

If the correlated noise is expressed as a covariance matrix $\mathbf{K}$, the likelihood takes the form:

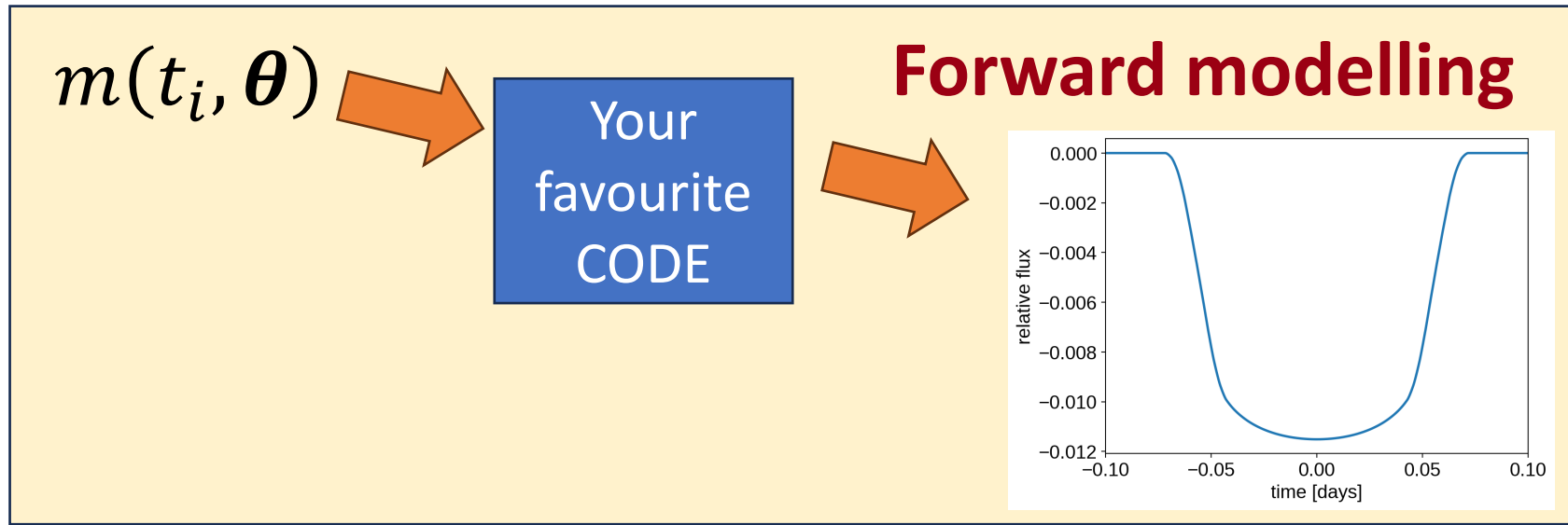
$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \frac{1}{\sqrt{|2\pi\mathbf{K}|}} \exp \left[-\frac{1}{2} (\mathbf{y} - \mathbf{m})^T \mathbf{K}^{-1} (\mathbf{y} - \mathbf{m}) \right]$$

Gaussian process regression: $\mathbf{K}$ is parametrized through a kernel

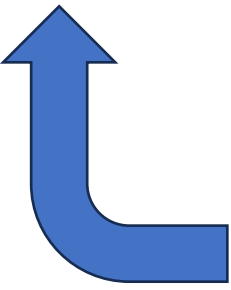
$$K[i, j] = \phi_1^2 \exp \left\{ -\frac{\sin^2[\pi(t_i - t_j)/\phi_2]}{2\phi_3^2} - \frac{(t_i - t_j)^2}{2\phi_4^2} \right\} + \delta_{i,j}\sigma_i^2$$



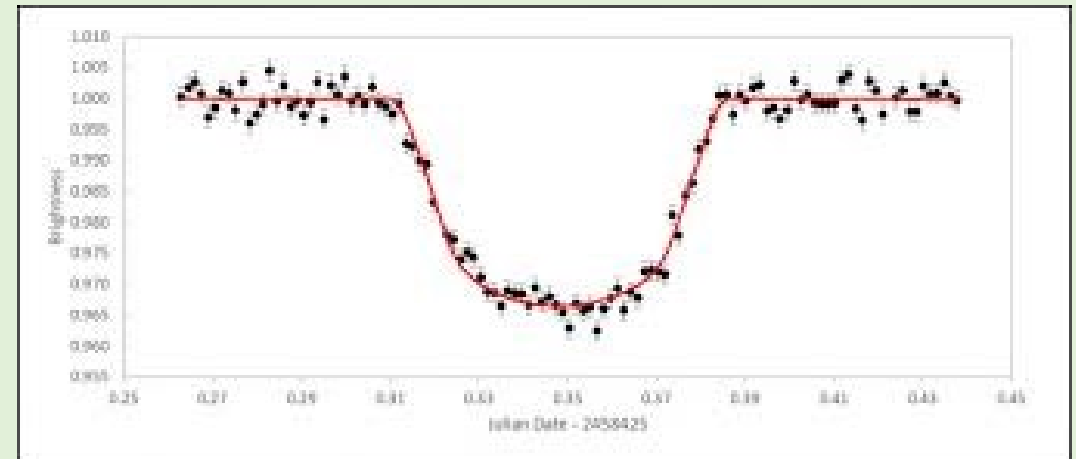
Forward modelling and Bayesian analysis



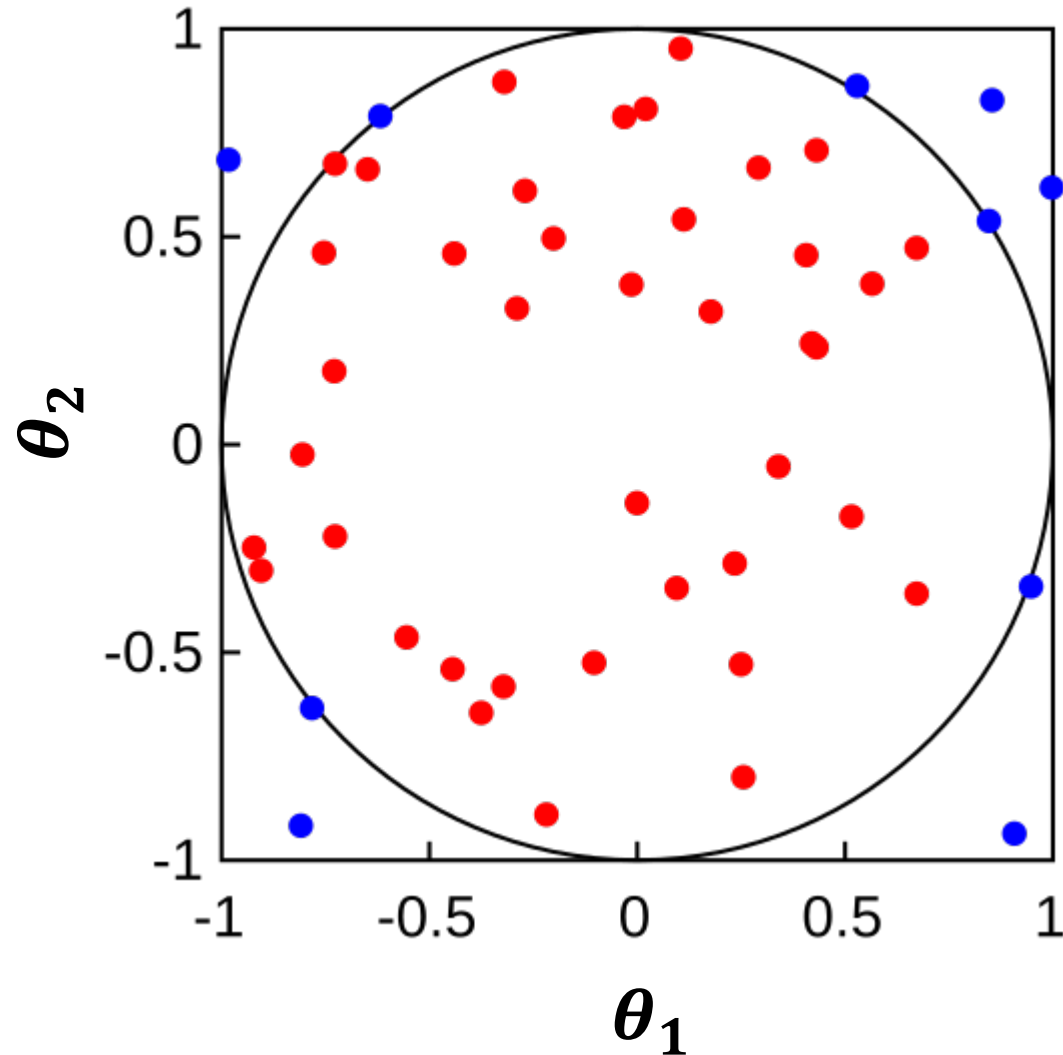
Generate new θ
to find $P(\theta|\mathbf{D})$



Compute log-probability $\ln(P(\theta|\mathbf{D}))$



Markov chain Monte Carlo

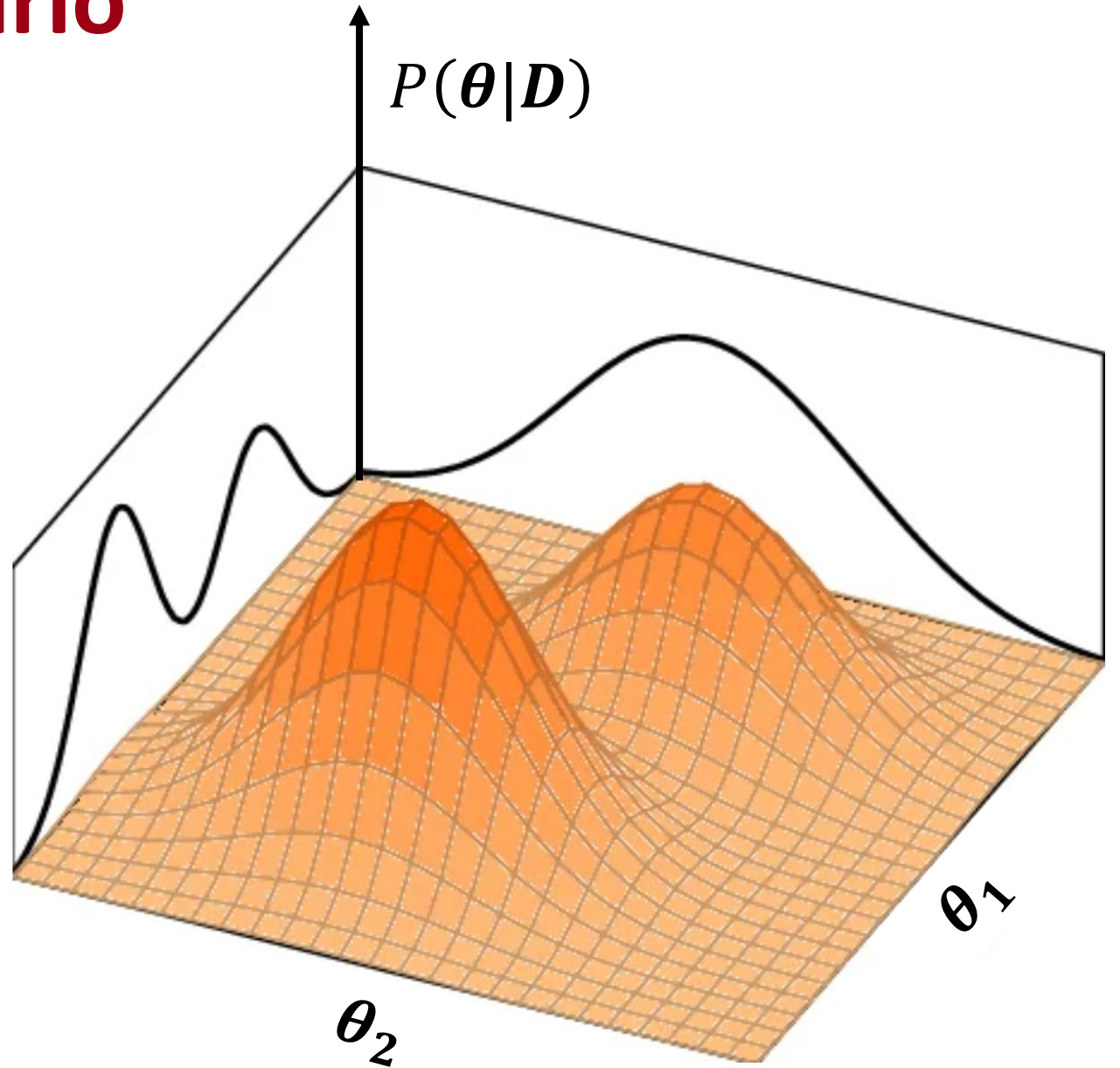
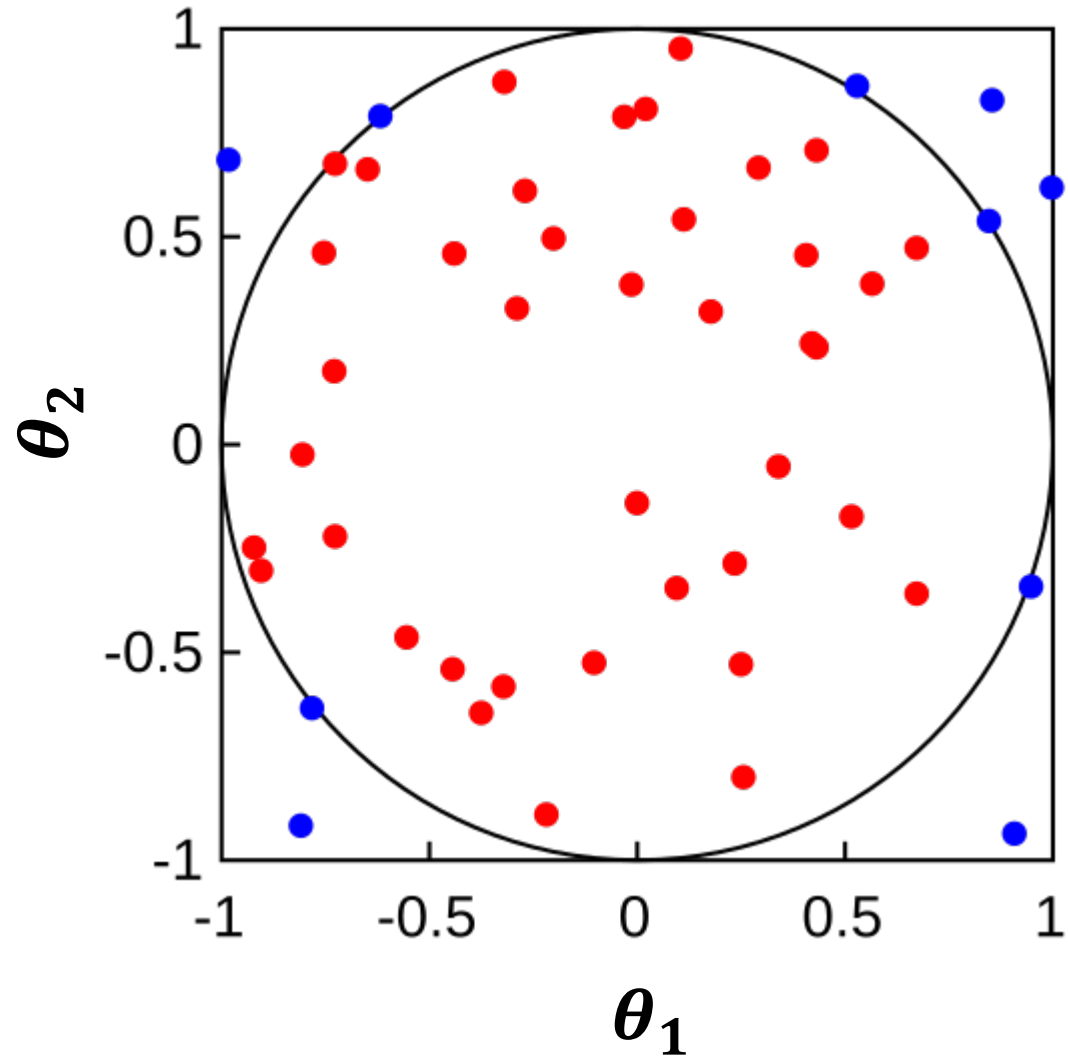


Monte Carlo integration: numerical integration using generation of random numbers

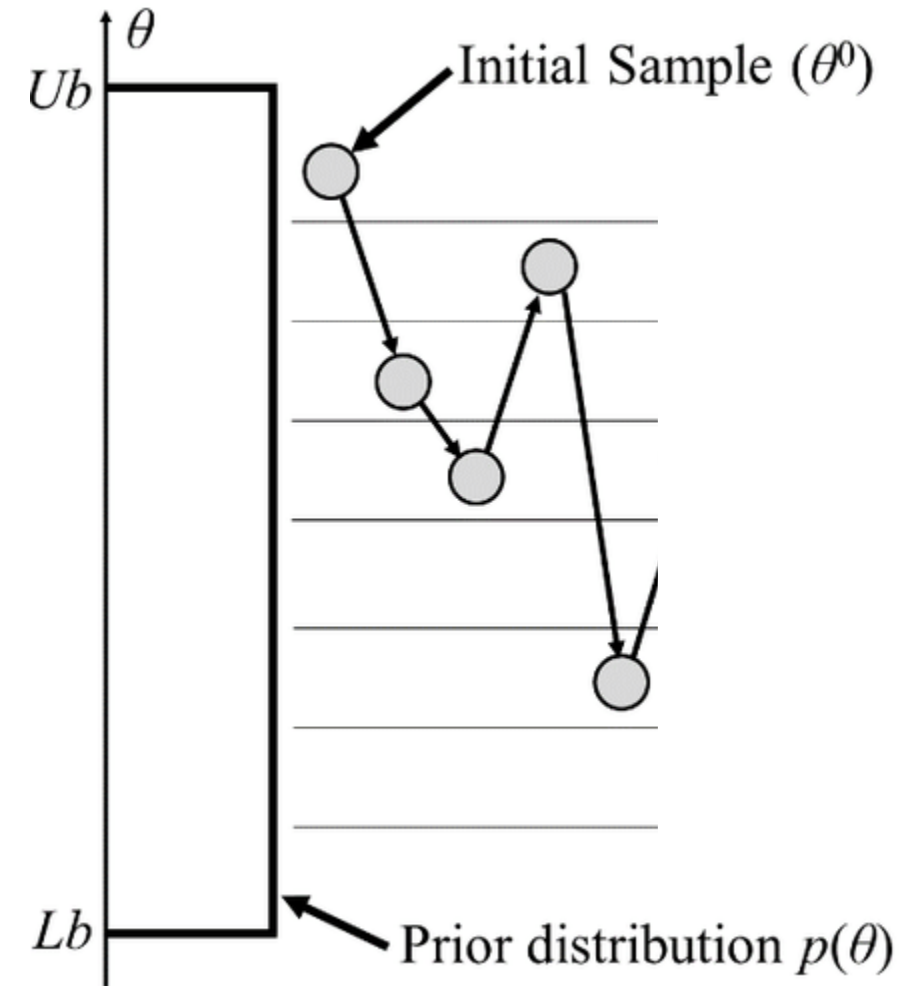
In the example, the area of the circle is computed by counting the number of random samples inside and outside the circle

This works well for high-dimensional problems or non-analytical shapes, but only if a high number of samples is employed

Markov chain Monte Carlo



Markov chain Monte Carlo

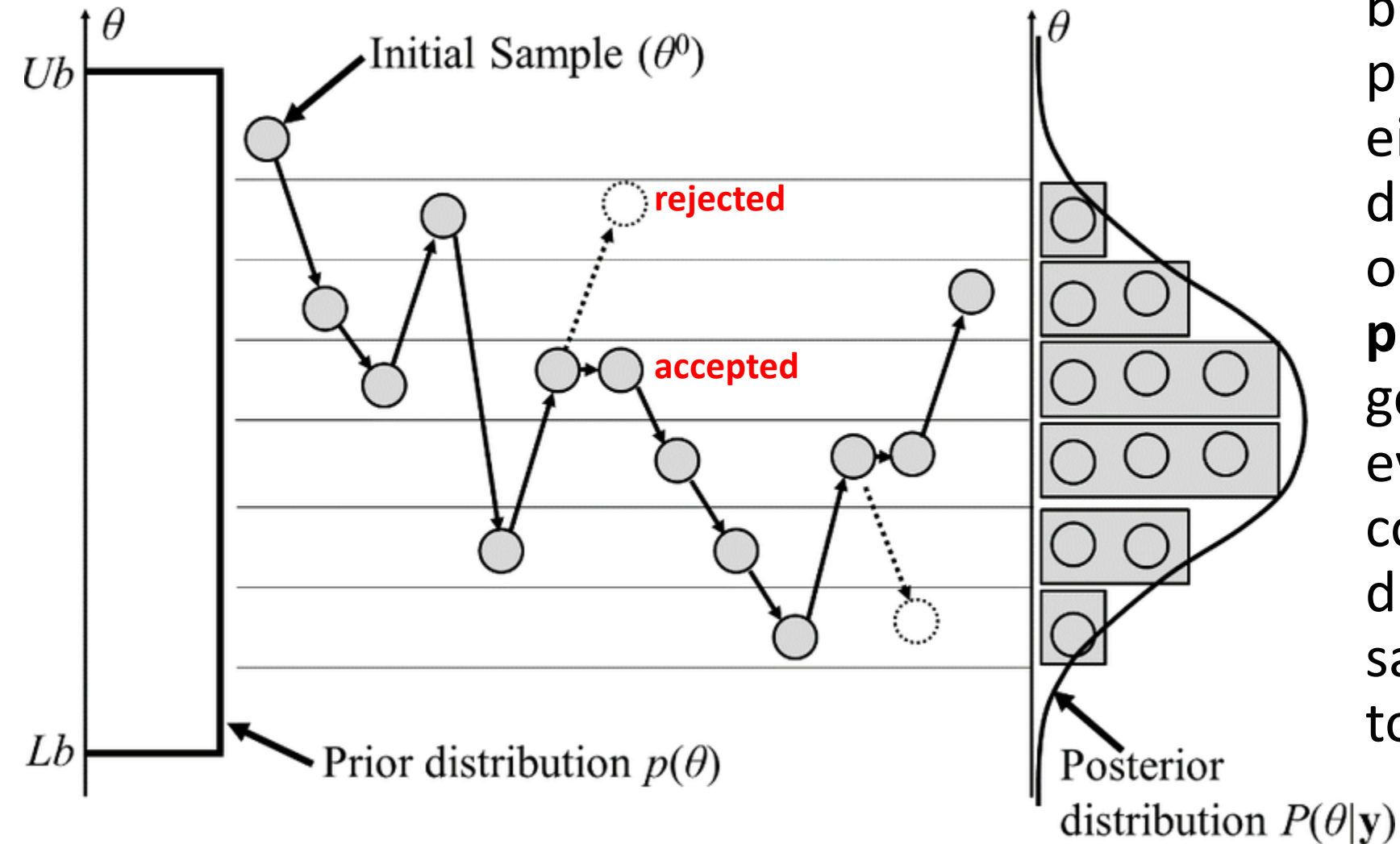


Markov chain: sequence of steps of possible values of θ in which the probability of each *step* depends only on the value of θ in the previous step

Each step of the chain is obtained by perturbing the previous value of θ . The amount of perturbation depends on the algorithm of choice (Gibbs, Metropolis-Hastings, Hamiltonian...)

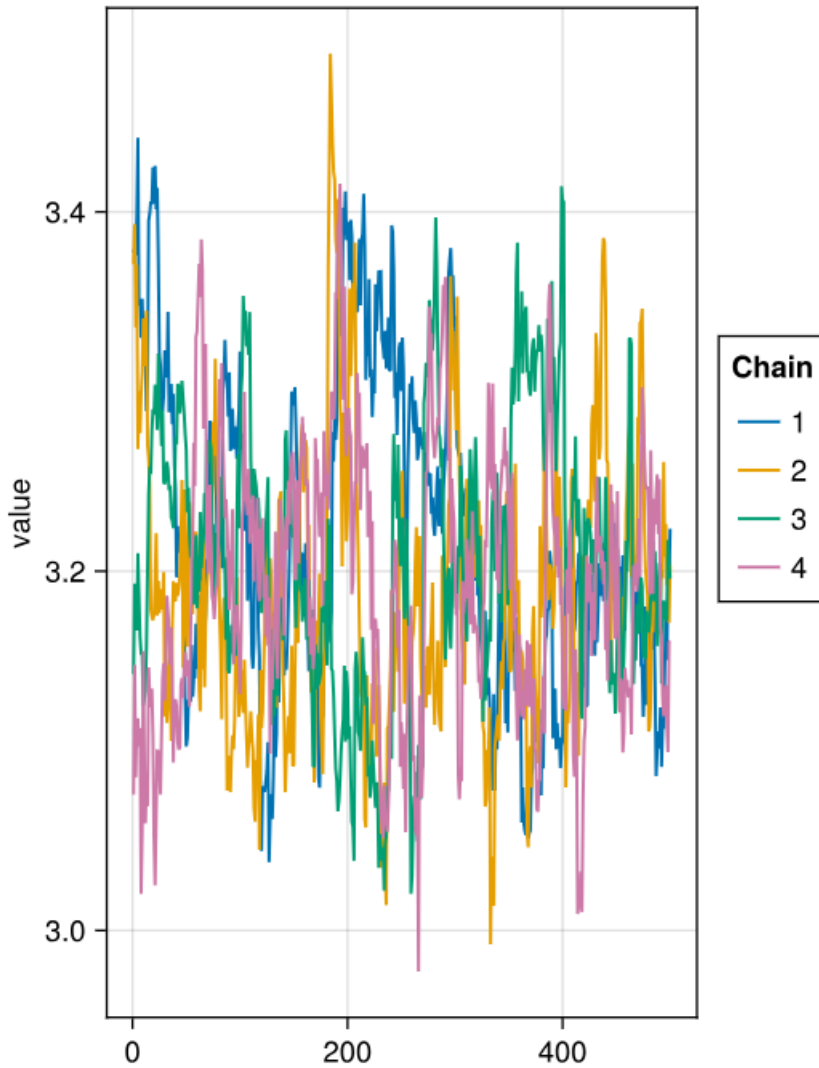
MCMC methods produce autocorrelated samples: successive draws from the Markov chain are statistically dependent.

Markov chain Monte Carlo



Not all the new steps will build the posterior: a proposed new state is either kept (**accepted**) or discarded (**rejected**) based on the **acceptance probability**, ensuring the generated samples eventually match the complex target probability distribution, with rejected samples allowing the chain to explore different regions

Autocorrelation of MCMC



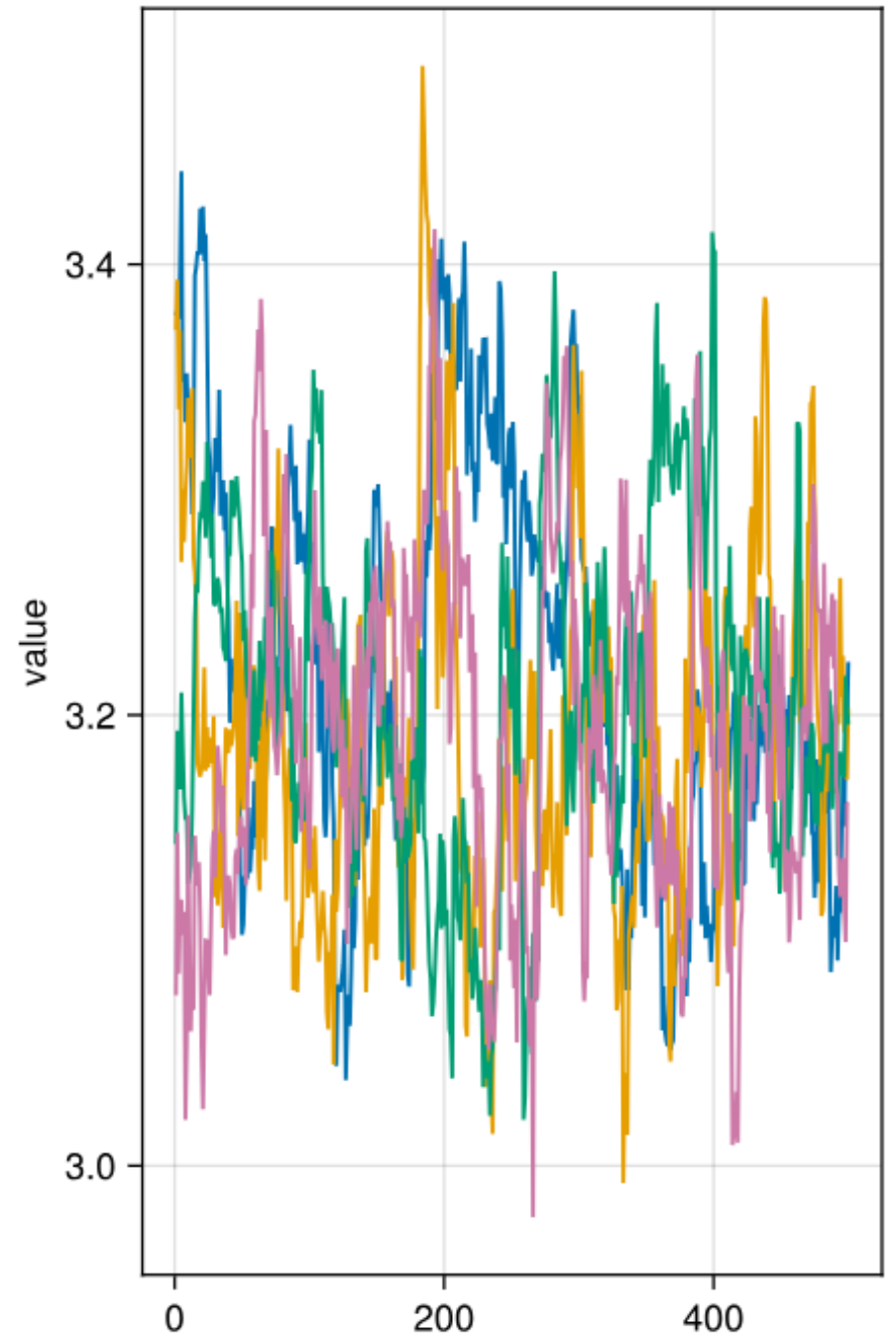
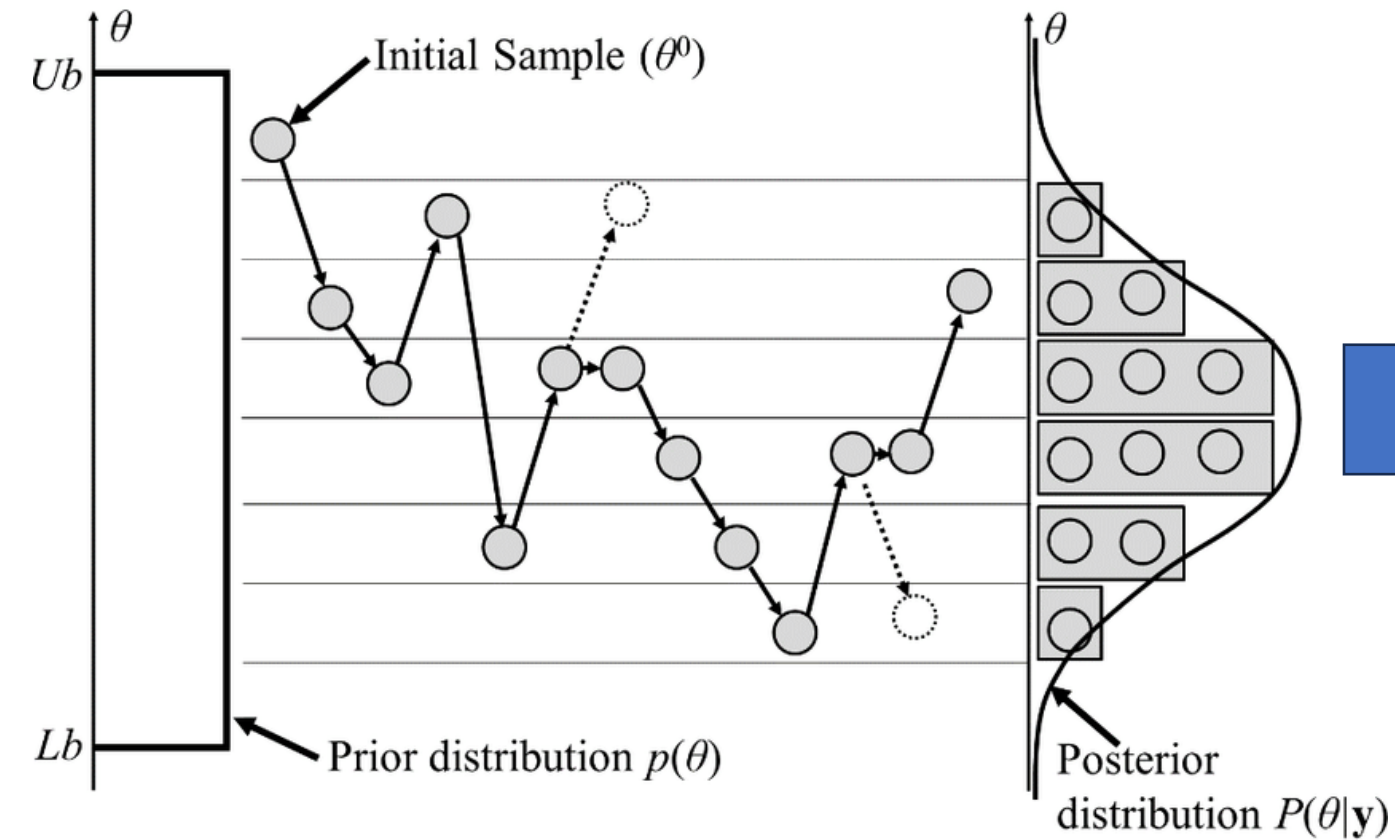
MCMC methods produce autocorrelated samples *within a given chain*

Each step only depends on the previous one: the correlation with step i and step $i + n$ decreases with increasing n

The **AutoCorrelation Function (ACF)** estimates how many steps it takes for a chain to produce two independent samples

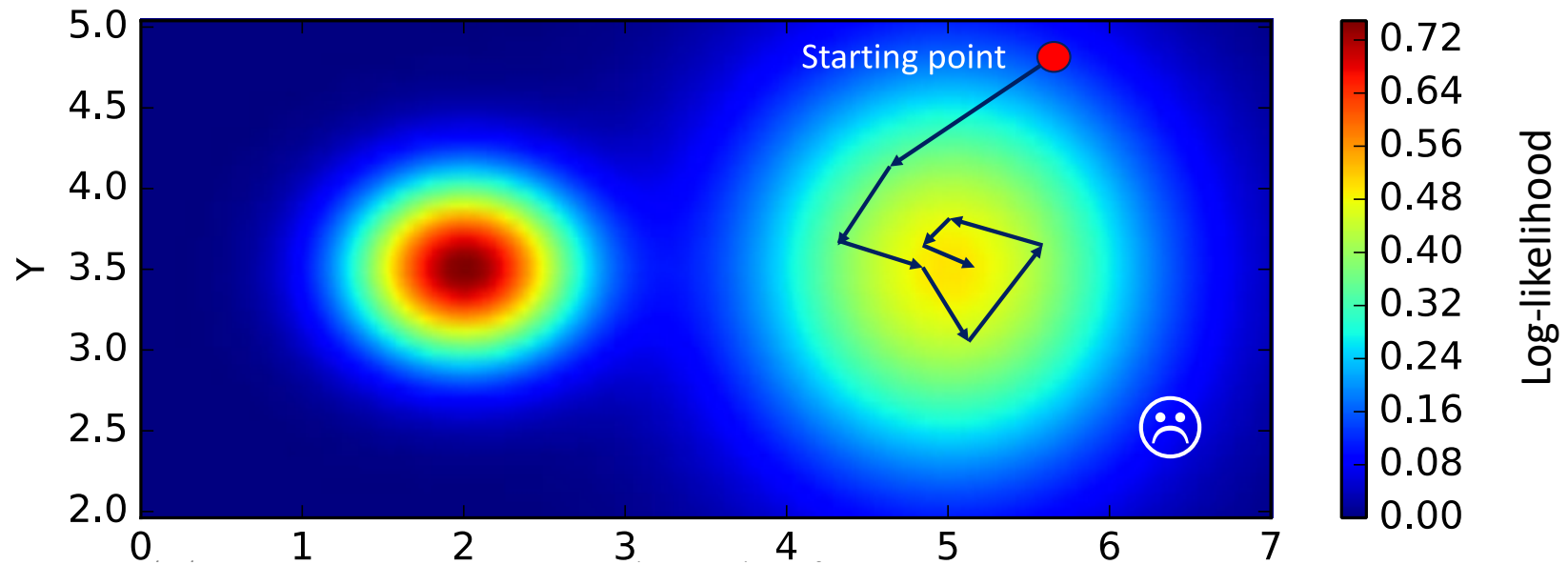
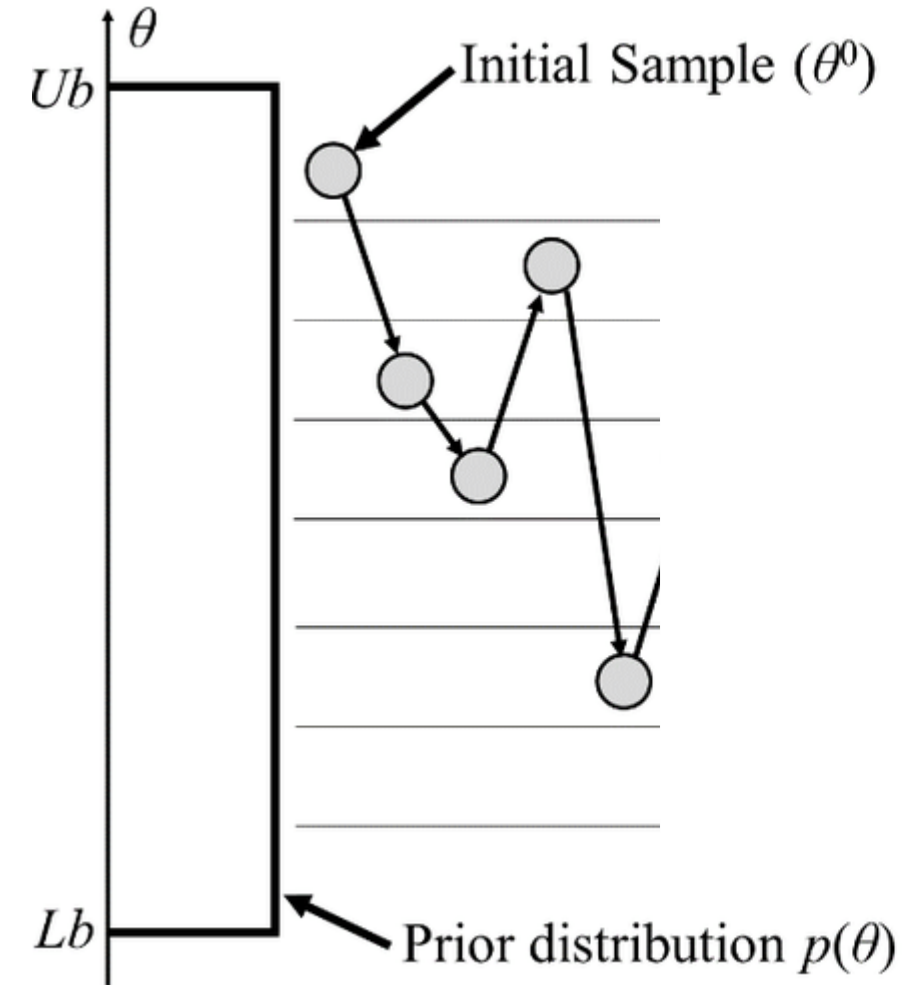
Thinning refers to the process of using only one step every N (*thinning factor*) to build the posterior. N should be equal or greater than the value provided by the ACF

Autocorrelation of MCMC



Thinning refers to the process of using only one step every N steps (*thinning factor*) to build the posterior

Starting point of the MCMC

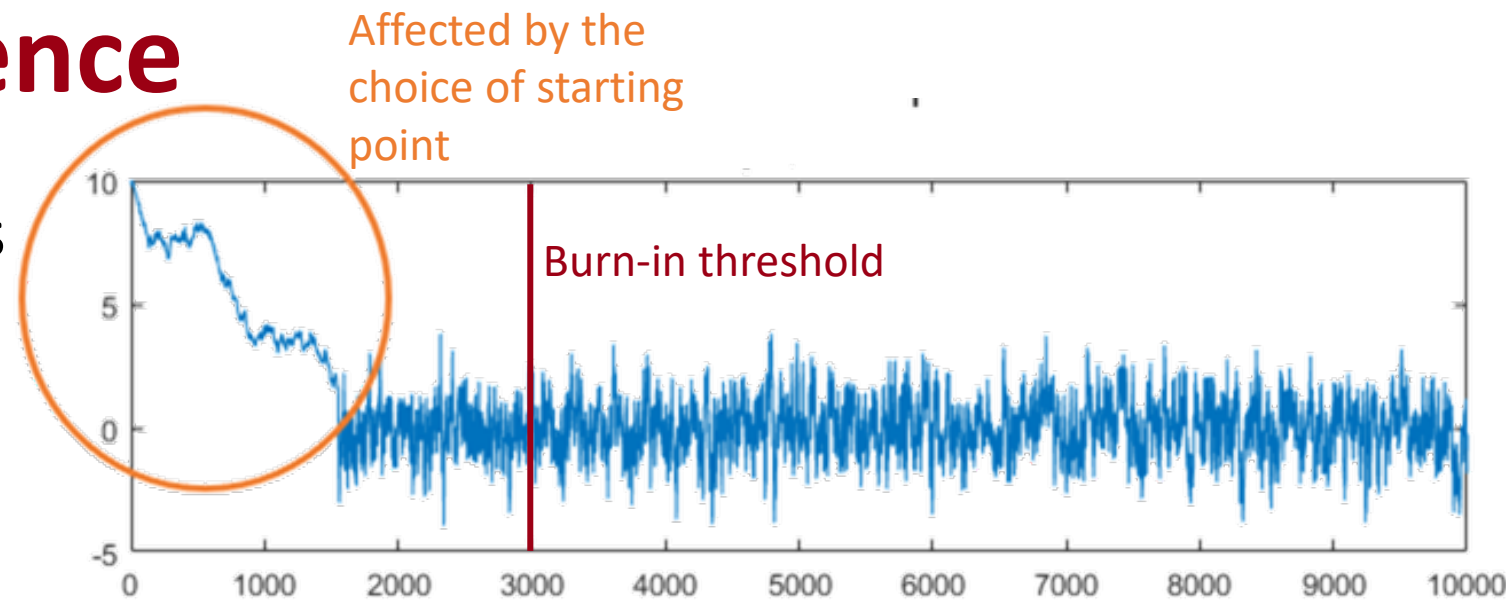


By construction, MCMC requires a **starting point**. The choice of the starting point is extremely important: the chains can get stuck in a local *maximum of the likelihood* (local minimum of the χ^2) as steps are correlated.

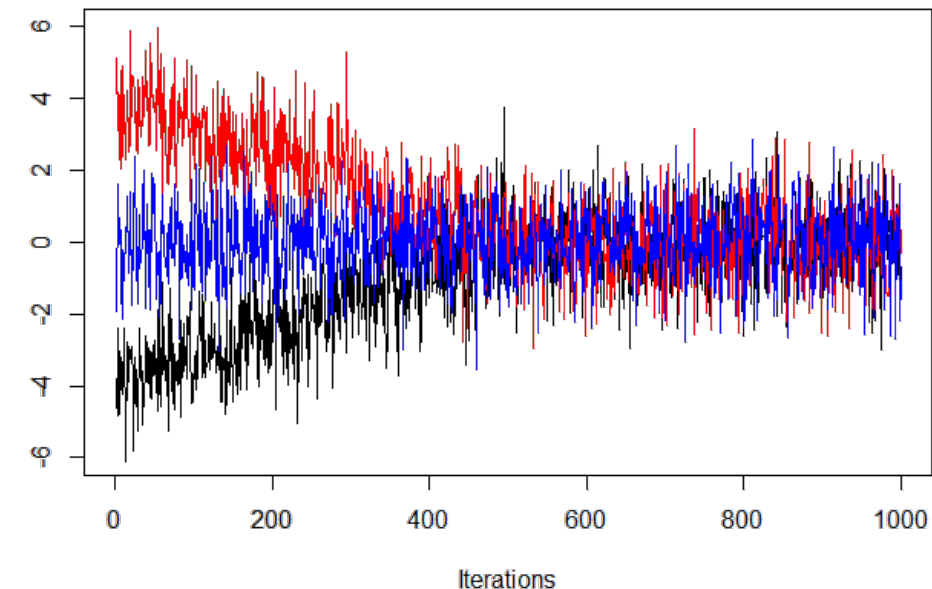
$$\ln \mathcal{L} = -0.5N \ln(2\pi) - \sum_{i=1}^N \ln(\sigma_i) - 0.5\chi^2$$

Burn-in and convergence

Burn-in phase: the initial samples may be affected by the choice of the starting point and should be rejected

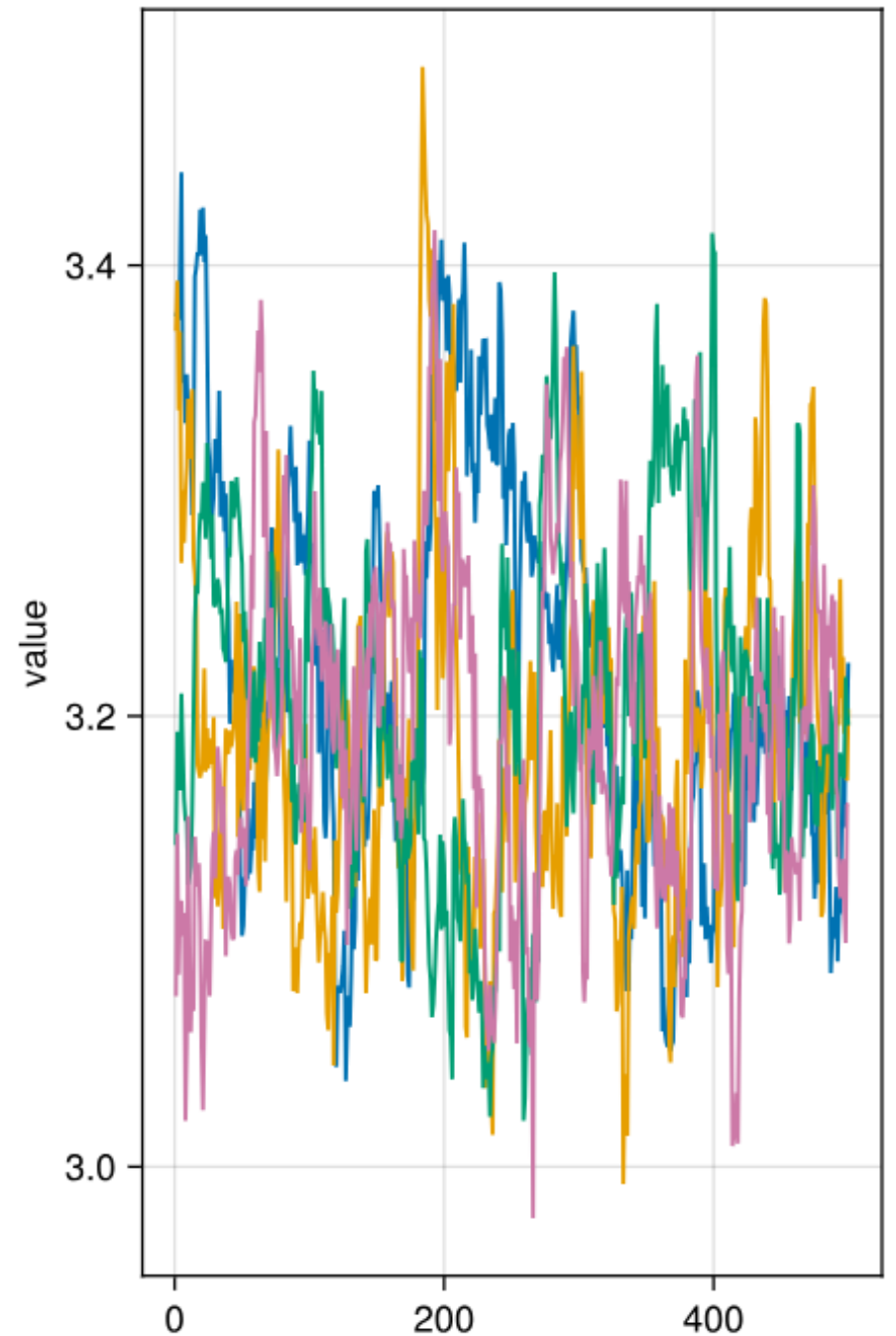
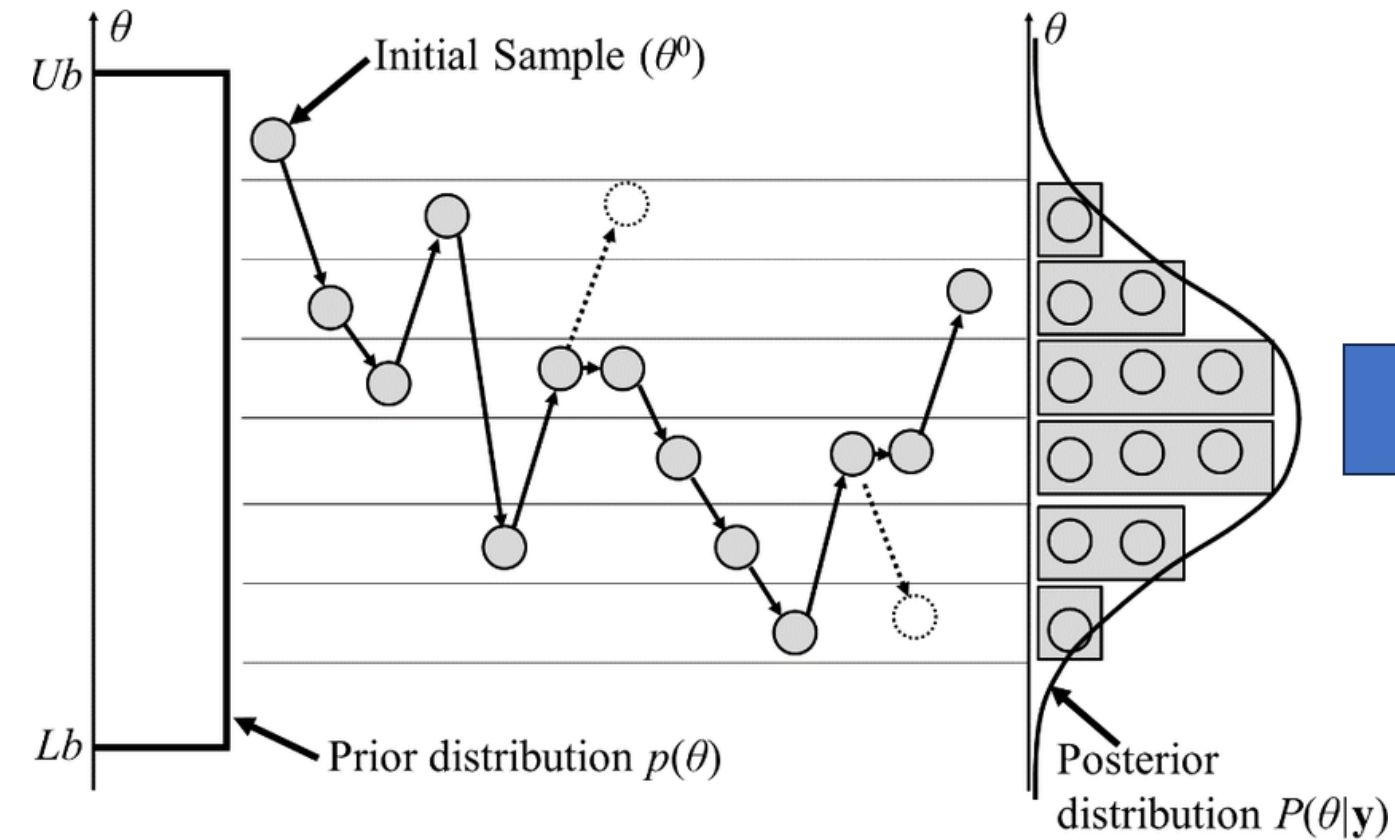


MCMC will **asymptotically converge to the target distribution**.
However, you cannot run your sampler forever... so, when to stop?



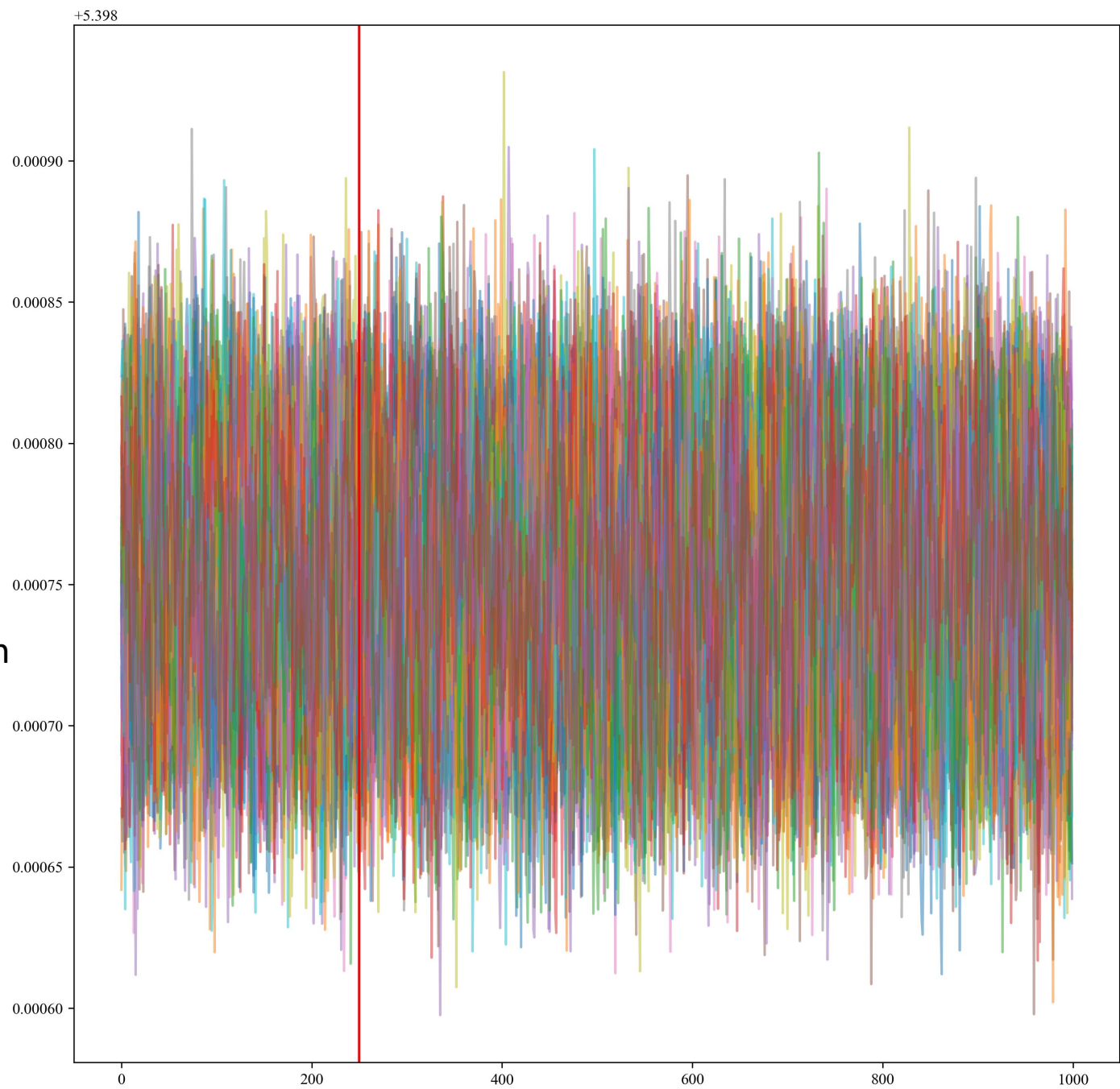
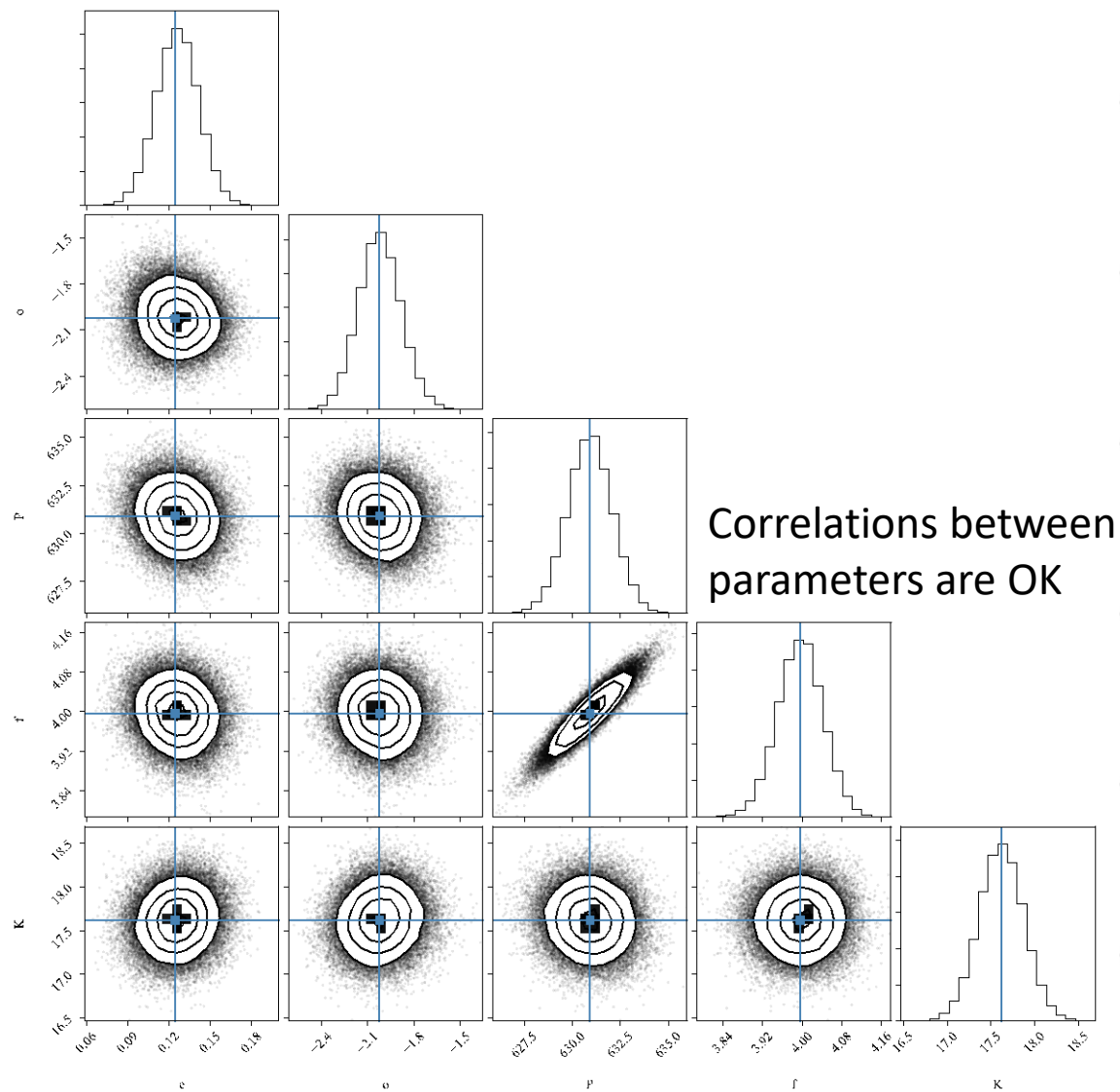
Convergence criterion: any of the moments (e.g. the mean or standard deviation) of the posterior distribution is independent of the choice of the steps or the chains used to build the posterior
E.g.: steps in the range [600-800] and [800-1000] will produce the same posterior distribution

Autocorrelation of MCMC



Thinning refers to the process of using only one step every N steps (*thinning factor*) to build the posterior

Example of a “good” chain



First step: identifying the model

Given the complexity of measuring the transit parameters in presence of limb darkening, the forward modelling approach is preferred.

P : orbital period

T_c : time of inferior conjunction

e : orbital eccentricity

ω : argument of periastron

$R_p/R_\star$: scaled planetary radius

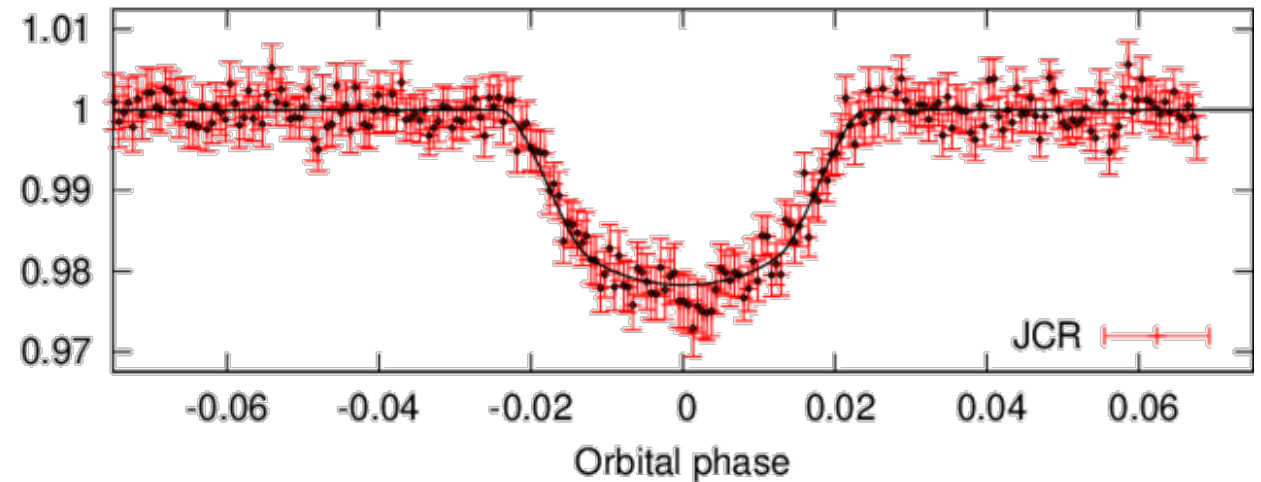
Either $a/R_\star$: scaled semi-major axis

or $\rho_\star$: stellar density

Either b : impact parameter

or i : orbital inclination

Other parameters, such as the temperature of star or day/night side of planet, may be required depending on the code



Incomplete list of transit modelling softwares:

BATMAN (Kreidberg 2015, [github](#))

PyTransit (Parviainen 2015, [github](#))

Transit and occultations:

SPIDERMAN (Louden & Kreidberg 2018, [github](#))

starry (Luger+2019, [github](#))

Detached eclipsing binaries and transiting planets:

JKTEBOP (Southworth 2012, [link](#))

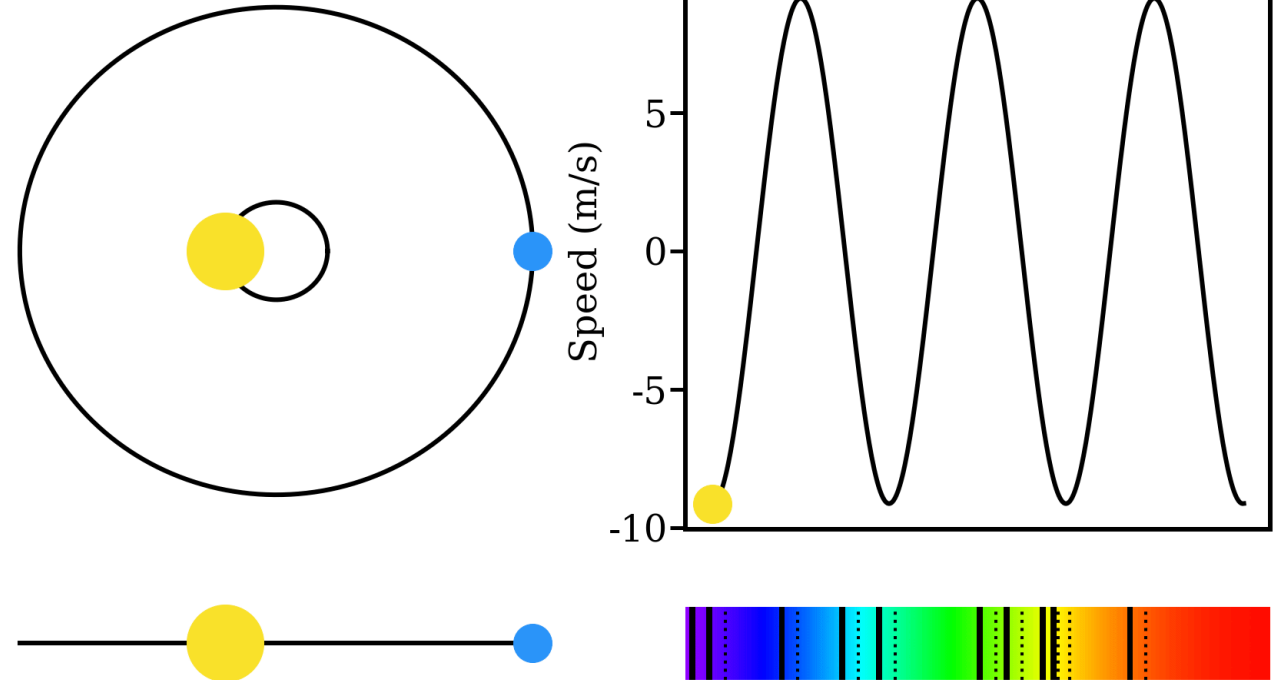
ellc (Maxted 2016, [github](#))

First step: identifying the model

Alysa Obertas (@AstroAlysa)

P : orbital period
 T_c : time of inferior conjunction
 e : orbital eccentricity
 ω : argument of periastron
 K_p : radial velocity semi-amplitude

$$RV(v) = -K_p(\cos(v + \omega) + e \cos \omega)$$



For a circular orbit:

$$RV(t) = -K_p \sin\left(2\pi \frac{t - T_c}{P}\right)$$

$$K = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{M_p \sin i}{(M_s + M_p)^{2/3}} \frac{1}{(1 - e^2)^{(1/2)}}$$

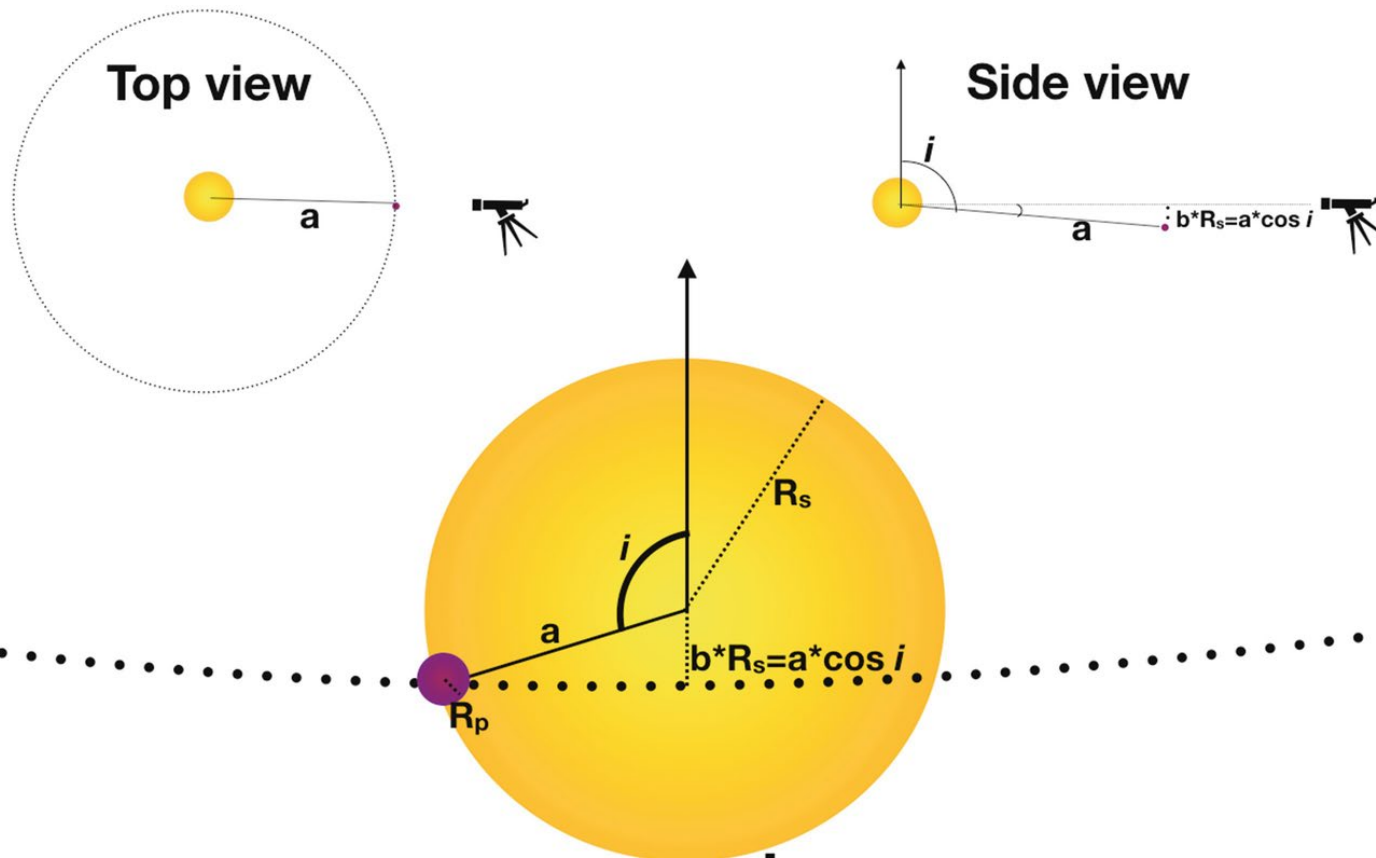
Change in parametrization

Seager and Mallen-Ornelas 2003

The Kepler Third Law can be used to obtain the **stellar density** as a function of a/R_* :

$$\frac{a^3}{P^2} = \frac{G(M_* + M_p)}{4\pi^2} \Rightarrow \rho_* + k^3 \rho_p = \frac{3\pi}{GP^2} \left(\frac{a}{R_*} \right)^3$$

Why: through isochrone fitting, the density can be estimated independently from the light curve



The **impact parameter** is the minimum projected distance of the planet from the star, in unit of stellar radii

$$b = \frac{a}{R_s} \cos i$$

Why: b is directly driving the transit shape

MCMC code

emcee: <https://emcee.readthedocs.io/en/stable/>

Nested sampling code

dynesty: <https://dynesty.readthedocs.io/en/latest/>

Corner plots

Corner: <https://corner.readthedocs.io/en/latest/>

pyGTC: <https://pygtc.readthedocs.io/en/latest/index.html>



**MCMC APPROVED BY
JUDGEMENTAL CATS**



**WAITING FOR MCMC
TO FINISH**